

About HELP Math

HELP Math Program Description

Overview

HELP (Help with English Language Proficiency) is a web-based mathematics intervention designed to address the specific needs of English language learners struggling to reach proficiency. For example, to understand mathematics, a student needs to be able to read, problem solve and communicate using technical language (e.g., *coefficient*, *hypotenuse*, *tessellation*) in a specialized context (e.g., *value*, *scale*, *chance*), with the mathematical definitions of these terms being much more specific and complex than their everyday definitions. Developed according to the principles of Sheltered English Instruction, HELP is effective for students in grades 3-8 who need to develop technical and academic English and prior math knowledge, in order to achieve proficiency. HELP is also used in high schools for remediation and as foundations for Algebra and Geometry. It is the only internet-delivered math program that scaffolds student learning by embedding research-based techniques directly into interactive, multimedia content.

Using Sheltered English Instruction means that math content and language are concurrently developed in meaningful contexts. HELP uses audio, video, text, and interactivity synchronized to create a visual connection between words, symbols, and meaning. For example, corresponding vocabulary and pictures flash in sync with audio. For additional support, all content is translated into Spanish.

Math is taught using clear explanations broken down into bite size steps. HELP screens are “clean” with few distracters, enabling students to maintain focus as they continually interact with real-world scenarios, games, and interactivities (number lines, base10 blocks). This varied and contextualized hands-on experience helps students learn math facts, skills, and concepts. HELP is comprehensive: 73 two-three hour self-paced lessons, over 200 hours of state and national standards-aligned math content, also organized in a learning object repository (accessible as standalone units of instruction). HELP topics have been chosen to make connections to the *NCTM* (National Council for Teachers of Mathematics) *Curriculum Focal Points* and state English Language Development standards. Problem-solving and test-taking strategies are explicitly taught.

Sections and Lessons

Since HELP is an activity-based, self-paced course with a modular structure, students can move around at their own rate. Each lesson follows the same logical structure with key pedagogical principles embedded in the instructional multimedia.

Real-World Scenario and Game: Real world math example to immerse the student in lesson content by connecting to students' prior knowledge.

Vocabulary: Vocabulary used in context, builds math fluency and prepares students to apply words to concepts.

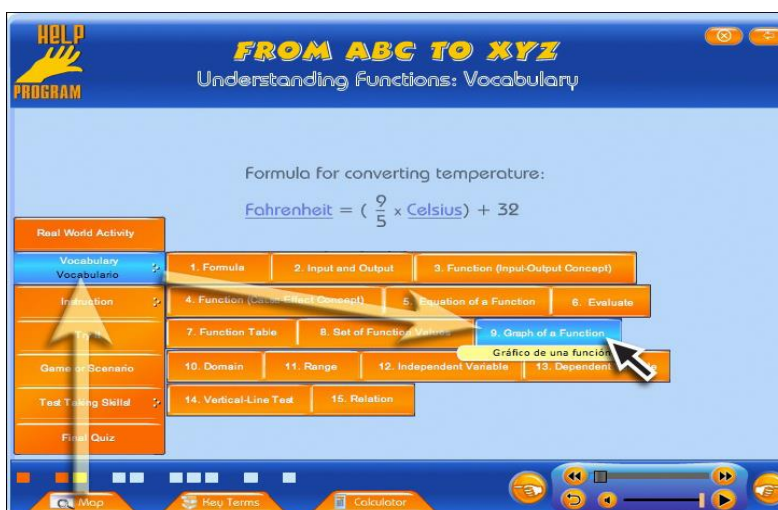
Instruction: Math concept and skills are developed interactively in the context of California math standards. Explicit instruction and models throughout with multiples opportunities for guided practice.

Try-It: Math skill practice using “bite-size” problem-solving questions with immediate feedback and error analysis.

Problem-Solving Skills: Students learn to read and comprehend assessment questions, using cognitive learning strategies and develop structured multi-step problem-solving strategies.

Final Quiz: Measures proficiency on content learned. Items are randomized and autoscored.

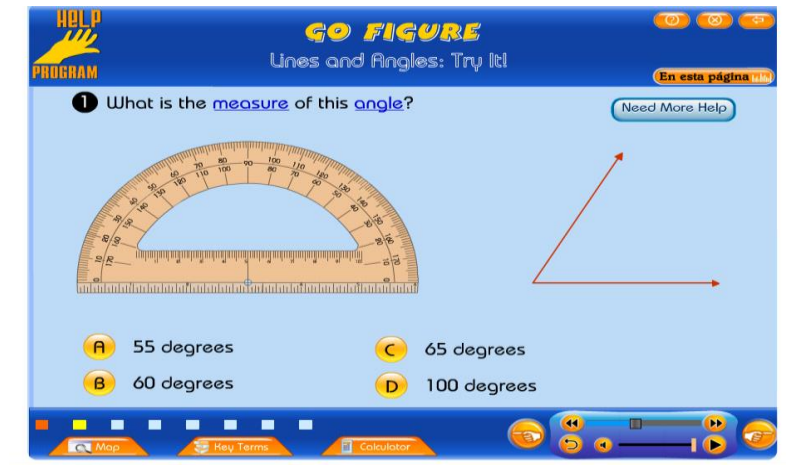
The menu is accessible from each instructional page. The user is able to navigate between sections and sub-sections in the program by clicking on pull-out menus, enabling them to go straight to the page they want.



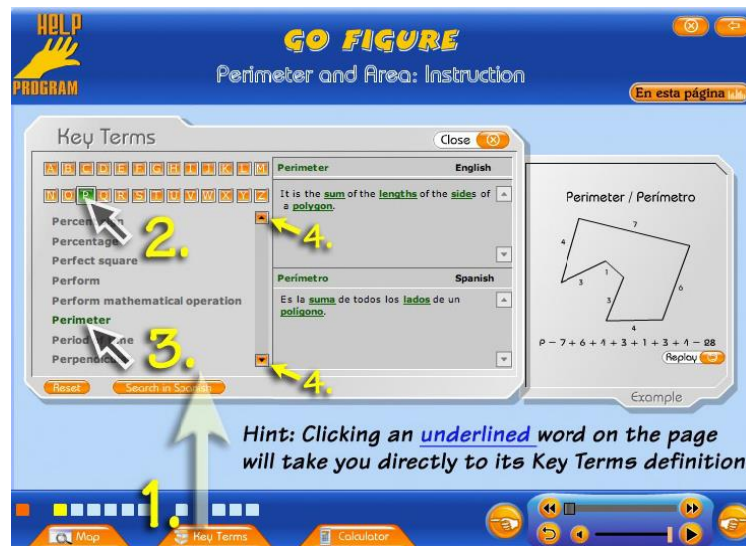
- ✓ Clicking a “Key Terms” button brings up a self-contained screen with all the math terms used in the lesson, presented in English, Spanish with a pictorial example.
- ✓ Clicking a “Calculator” button makes the built-in calculator appear.
- ✓ Clicking the “En Esta Pagina” button brings up a Spanish translation of the instructional content on the page.

The content encourages skill development

Math skill development is explicitly addressed in the Instruction and Try It sections of each lesson. The interactivity throughout the program, offers the student multiple opportunities to develop and practice their skills, with support and extra help available as needed.

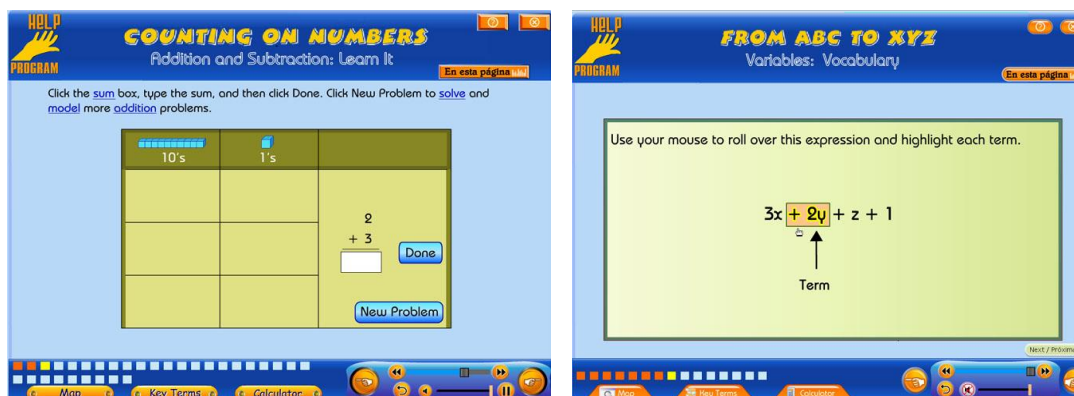


The content also focuses on the technical vocabulary and academic language needed to succeed in math. The support is available in context through hyperlinks and through a comprehensive key terms dictionary. Searches can be done in English or Spanish and there is always a picture example.



Interactivity throughout the program

There are interactivities and “manipulatives” throughout the HELP program. HELP activities are used to support the learning of concepts as well as language.



Students Help Themselves

Supportive instructional tools may be accessed by students any time during the lesson:

- *Key Terms Dictionary & Hyperlinks*: definitions and pictorial examples in English and Spanish.
- *Hints/Need More HELP Buttons*: constructive and immediate feedback
- *Calculator*
- *Manipulatives*: protractors, fraction strips, thermometers, etc.
- *Spanish audio*: complete translation for all content
- *Social network tools*: email teacher with queries

A summary of HELP features:

- ✓ HELP is student-centered and allows students to engage with material at their own pace. On the computer, students can move faster or slower through a lesson depending on their ability.
- ✓ HELP courses are highly interactive, colorful and musical. Technology allows the instruction to trigger multiple modalities, a major benefit when teaching ELLs. HELP's modules incorporate voice, music, visuals, text, and interactivity in order to keep students engaged in ways that text books or lectures cannot.
- ✓ Vocabulary development is targeted emphasizing math content language as well as academic English; available to students within the narrative presented to them (as hyperlinked words) as well as within an online dictionary of mathematical terms with bilingual support available.
- ✓ Comprehensibility is enhanced by scaffolding the learning, e.g. breaking down concepts into small, comprehensible "learning chunks" and synchronizing visuals, audio and interactivity.
- ✓ Instructional support in critical areas is available in Spanish. Technology allows, promotes, and supports individualized learning, where each in-class HELP user receives targeted instruction that meets their individual needs.

- ✓ Non-graded assessment and interactive feedback loops are designed to provide students with constructive feedback and an opportunity to learn from their mistakes.
- ✓ Students are taught test-taking skills – how to read, comprehend and answer ‘look alike’ questions from state standardized math tests.

Teacher Support Tools

Teachers are provided with a wide-selection of guides. Teachers can access and present student guides (quick start and full length) and tutorials (Flash movies and PDFs), provided in English and Spanish. There are HELP content alignment tables which correlate to state and NCTM standards, Teacher Administration Guides (PowerPoint, PDF), a Guide to the HELP Methodology including implementation strategies, as well as pre-recorded Webinars to enable teachers to review the various HELP support tools and features. There is also a comprehensive Teacher’s Guide with suggested activities for concrete and verbal practice, and/or exploratory, hands-on activities.

Teachers are able to track progress in real-time and monitor student time spent on the program and auto-generate computer reports based on quiz scores. Teachers also have access to a large number of support tools and features, which enable them to control the HELP Math experience for an individual student or whole class. Custom features include the ability to turn on/off linear content navigation, Spanish audio, calculator (during lessons and/or quizzes), as well as to set content mastery levels and allow students to email queries which may arise. Teachers have the option of ‘locking down’ the pathway through a math lesson, thus forcing the student to work through the content sequentially (e.g., following HELP’s structured lesson design from Real World Scenario to the Final Quiz sections, detailed above).

HELP lessons are also organized as grade-level learning objects. There are approximately 6,500 instructional swf pages meta-tagged and accessible in a learning object repository. Teachers can use these objects to create custom lessons designed around a particular learning objective. Alternatively, they can shorten a lesson or present only certain sections in a whole class exercise. A teacher can also create a custom curriculum. For example, a teacher may require a particular student(s) to work on content from earlier grades to build prior knowledge before accessing grade-level lessons.

HELP’s Diagnostic pretest / and Automatic Custom Curriculum Assignment feature builds on the University of Colorado’s research-based and implemented diagnostic pre-post tests, which are available to schools through the HELP program. HELP contains hundreds of diagnostic assessments that are aligned to both independent academic achievement standards and HELP content. This enables the program to prescribe content and automatically place the student in the part of the program they require. After the student takes the appropriate grade-level test, it is auto-scored and a diagnostic report is auto-generated, showing student content, skill, and prior knowledge and skill gaps. These gaps are matched to HELP lessons—with a push of a button, a teacher can auto-generate a custom curriculum and unique learning pathway for each individual student in their class. Below is a sample of one missed question.

Question 1

SKILL MISSED:

Number, operation, and quantitative reasoning.

The student adds, subtracts, multiplies, and divides to solve problems and justify solutions.
The student is expected to: use multiplication and division of whole numbers to solve problems including situations involving equivalent ratios and rates.

BACKGROUND KNOWLEDGE MISSED:

[Math Foundations 1 Lesson: 4 \(Division\)](#) - Dividing a multi-digit number evenly by a one-digit divisor

[Math Foundations 2 Lesson: 6 \(Division\)](#) - Dividing multi-digit numbers by one-digit divisors using models (Hundreds Chart with Skip Counting, Arrays, and Checking Long Division with Multiplication)

[Math Foundations 3 Lesson: 6 \(Division Skills\)](#) - Performing long division with multi-digit divisors, using long division in problem solving situations, and checking quotients with multiplication and models

GRADE LEVEL LESSONS MISSED:

[Numbers Make Sense Lesson: 1 \(Positive & Negative Numbers\)](#)

Assign: Grade Level Lessons

Assign all: Background and Grade Level

A summary of HELP features is provided below:

- ✓ Auto scoring of student quizzes and randomization of test questions through DDI's proprietary Teacher Support & Administration system.
- ✓ Teachers are able to track progress in real-time and monitor student time spent on task through DDI's proprietary Teacher Support & Administration system.
- ✓ Auto generation of teacher reports (i.e. lessons completed, quiz scores, etc.) through DDI's proprietary Teacher Support & Administration system.
- ✓ Teachers can create a custom curriculum to target learning of particular concepts and differentiate the learning path for individual students.
- ✓ Teachers can set mastery levels for student achievement, thus ensuring that a student learns particular concepts before they move on to new content.
- ✓ A Diagnostic pre-test with an auto-generated custom curriculum. With a push of a button, a teacher can auto-generate a custom curriculum and unique learning pathway for each individual student in his/her middle school class
- ✓ Teachers can turn on/off features of the program such as the Spanish or calculator.
- ✓ Teachers have the option of 'locking down' the pathway through a math lesson, forcing the student to work through the content.
- ✓ The curriculum is delivered online so the material can be rapidly updated to reflect field research and evaluation results.
- ✓ Lessons align directly to National Council of Teachers of Mathematics (NCTM) Standards and to state content standards and assessment frameworks for mathematics (grades 6-8) and emphasize the NCTM Curriculum Focal Points.
- ✓ Teachers' Guide includes information on lesson alignment to standards, alignment to English language acquisition best practice principles, use of the program in a variety of settings, plus an online tutorial explaining product navigation and functionality (English and Spanish).

HELP Passwords

HELP is cross-platform, aligns to any learning management system, and works on any web browser. HELP lessons are password-based and delivered online. Students can access HELP in class, lab, library, community center, at home, or wherever they have access to a computer with broadband. Professional development is available to help teachers integrate HELP into their classrooms.

Research shows *HELP Math raises Students' Achievement*

In 2005, the Colorado Department of Education sponsored third party formative research, which resulted in a published White Paper. The HELP Program study results demonstrated 70% learning gains in math in the target population and 40% gains when research was extended to include high school students. In addition, significant and notable gains in English language proficiency were reported as an ancillary benefit. As a result of its early success, the HELP Program partnership including Rocky Mountain PBS, University of Colorado, Colorado Springs (UCCS), Colorado Department of Education, and DDI was awarded a *Ready to Teach* grant from the U.S. Department of Education in the amount of \$5 million (in its 4th and final year).

As part the U.S. Department of Education *Ready to Teach* grant, the University of Colorado at Colorado Springs undertook a large-scale, multi-year experimental study involving more than 1,200 students nationwide. Dr. Lindy Crawford, Associate Dean of the College of Education, is chief researcher of the grant. The two figures below show the student demographics by ELL condition and state.

Figure 1:

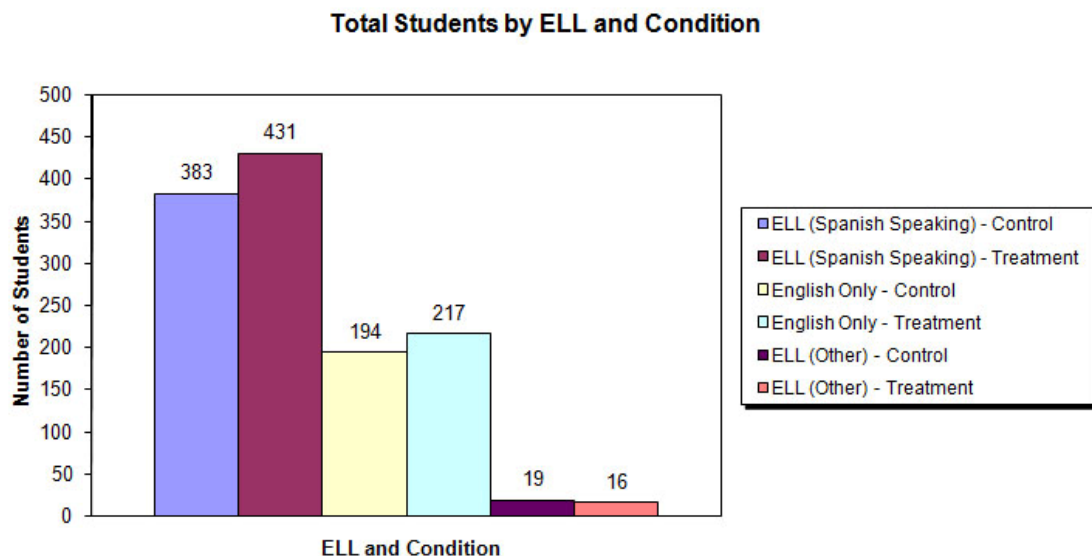
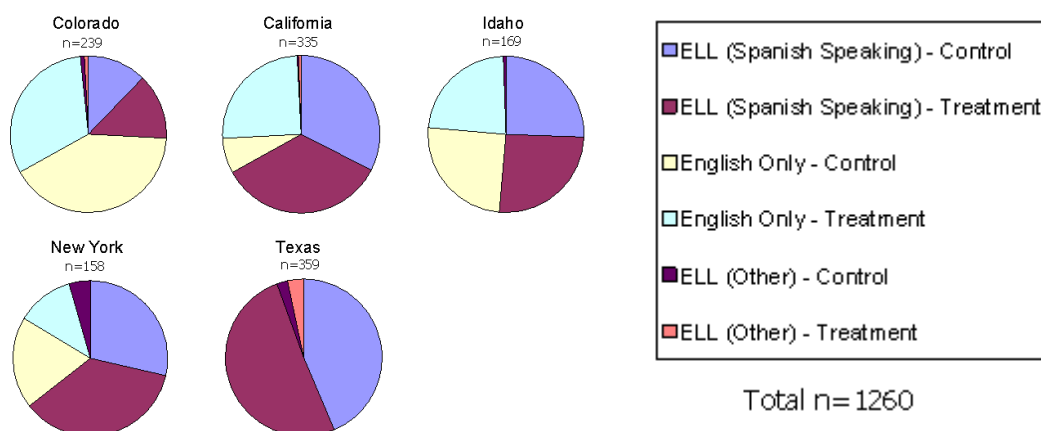


Figure 2:

Number of Students per State by English Language Learner Grouping



University of Colorado created a pre/post-test covering: Numbers and Operations, Algebra, Geometry, and Statistics, as well as a Basic Skills module. Reliable and valid tests targeted for use by middle school, Spanish speaking English language learners were developed and piloted in the summer preceding the 2007 academic year. Assessment design rigorously adhered to guidelines which aimed to make the test as fair and equitable as possible for all test-takers.

Dr. Crawford reports that students using HELP Math have shown significant increases on post-test math scores. Students targeted by HELP at language level 1 (least proficient in English) and level 2 (limited English proficient) demonstrated higher scores using HELP Math than students in the comparison group; gains were also shown by students with learning disabilities, who used the HELP intervention (Crawford, 2009). Additionally, calculations of AYP (adequate yearly progress) indicate an increase of 18% in the AYP in math for LEP/ELL students at the participating schools and all ELL students who used HELP Math in the research study showed increases in their state English language proficiency test from the previous year (Lawyer-Brook, 2008). Furthermore, teacher feedback on the program, in the form of surveys and focus groups, has been extremely positive (Lawyer-Brook). Quantitative and qualitative information from field studies has informed program modifications; with a strong focus on the needs of teachers and the factors involved in successful implementation, and sustainable use of a technology-based math intervention.

Significantly, researchers and teachers using HELP have found that; with its scaffolded instruction and focus on the intersection of math concept and skill development and English language acquisition, the program works extremely well with students with learning disabilities and *all* students needing extra support. For example, pre-post scores of Special Education students using in HELP Math in the University of Colorado Research Study showed mean gains of 15% (Crawford, 2008).

HELP Math is Award-Winning

For three consecutive years—2007, 2008, and 2009—HELP was “deemed to be of high quality” by the U.S. Department of Education’s (DOE) Government Performance and Results (GPRA) Act. The independent review panel of qualified experts, chosen by the U.S. DOE’s Office of Innovation and Improvement, gave HELP Math the highest rating of all technology-based

programs. The GPRA evaluation is an extremely rigorous one. The DOE sets standards and uses five categories to assess all products to be reviewed: Content, Technology, Design, Implementation, and Target Audience. The Independent Review panel members are given access to the complete HELP Program throughout the evaluation period. The external reviewers rate the product and provide evaluations for each metric. The past two years, HELP Math content has received an average score of 92 out of a possible 100.

HELP won the Software and Information Industry's (SIIA) 2009 CODiE Award for Best Instructional Solution for Special Needs Students. It was also selected as a finalist for Best Mathematics Solution. The winners were selected from a field of 850 nominations. The SIIA CODiE Awards is the only peer-reviewed program in the software and content industry.



**2009 SIIA CODiE Winner as
*Best Instructional Solution
for Special Needs Students***

